

CHARTRES CHAMPOL AP, France

WMO: 071430

Lat: 48.4603N

Lon: 1.5011E

Elev: 512

StdP: 14.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.3	25.6	16.4	12.9	26.1	20.5	15.6	27.9	23.0	47.8	20.4	48.0	5.4	10	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	88.0	67.4	84.0	66.2	80.3	64.8	69.7	83.3	67.9	80.0	66.3	77.1	7.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	95.7	73.7	63.7	90.0	72.1	62.1	84.9	70.4	34.0	83.5	32.5	80.2	31.2	77.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	15.9	14.1	16.0	94.7	6.9	4.5	11.0	97.9	7.0	100.5	3.1	103.0	-1.9	106.2
			15.3	72.7	6.8	2.0	10.4	74.2	6.4	75.3	2.6	76.4	-2.4	77.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	39.4	40.6	45.6	50.2	56.5	62.6	66.7	66.3	60.2	53.8	45.6	40.4	(6)
(7)		DBStd	11.46	7.29	6.75	5.87	5.97	5.70	5.63	5.39	5.03	6.08	6.05	7.05	(7)	
(8)		HDD50	1325	330	266	160	67	10	0	0	0	32	156	304	(8)	
(9)		HDD65	4846	792	683	602	443	269	111	46	45	161	347	583	764	(9)
(10)		CDD50	2203	3	2	23	74	212	379	519	507	305	151	23	5	(10)
(11)		CDD65	248	0	0	0	0	6	40	99	87	15	1	0	0	(11)
(12)		CDH74	2722	0	0	0	17	85	448	1043	928	189	12	0	0	(12)
(13)		CDH80	881	0	0	0	0	5	124	361	347	43	1	0	0	(13)
(14)	Wind	WSAvg	7.0	7.9	7.8	7.5	7.1	6.8	6.5	6.1	6.2	6.6	7.1	7.6	(14)	
(15)	Precipitation	PrecAvg	24.4	2.0	1.7	1.7	1.9	2.2	2.0	2.1	1.9	1.8	2.4	2.2	2.3	(15)
(16)		PrecMax	33.7	4.1	3.7	4.8	4.9	5.4	4.1	6.4	4.1	4.6	5.5	4.1	5.6	(16)
(17)		PrecMin	16.9	0.3	0.1	0.3	0.2	0.4	0.2	0.7	0.4	0.2	0.7	0.7	0.6	(17)
(18)		PrecStd	4.1	0.9	0.9	1.1	1.2	1.1	0.9	1.3	1.1	1.2	0.9	1.1	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	59.7	68.2	76.5	80.2	89.5	93.5	96.2	85.0	75.3	62.0	56.6	(19)
(20)			MCWB	52.8	50.6	54.0	60.4	66.0	70.5	69.3	69.9	64.8	63.0	58.4	53.9	(20)
(21)		2%	DB	53.6	54.5	63.1	71.3	76.4	83.7	87.8	88.4	79.6	69.9	58.9	54.4	(21)
(22)			MCWB	51.1	49.7	52.6	58.3	62.7	67.3	67.1	67.0	63.5	61.2	55.9	52.1	(22)
(23)		5%	DB	51.4	52.1	58.8	66.2	72.4	79.0	83.8	83.4	74.8	65.7	56.4	52.5	(23)
(24)			MCWB	49.1	48.6	51.1	55.1	60.8	65.3	66.1	65.9	61.9	58.9	53.6	50.4	(24)
(25)		10%	DB	49.2	50.1	55.4	61.8	68.5	75.2	80.0	78.9	71.0	62.6	54.2	50.4	(25)
(26)			MCWB	47.0	47.2	49.9	52.6	58.6	63.2	64.8	64.2	60.2	57.5	51.8	48.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	52.6	56.6	62.3	67.9	72.4	71.0	71.8	67.7	64.5	59.6	54.7	(27)
(28)			MCDWB	55.3	56.2	64.5	73.5	76.9	84.5	87.6	89.2	79.3	71.4	61.4	56.0	(28)
(29)		2%	WB	51.5	50.7	54.0	58.7	64.5	69.0	69.2	69.4	65.3	62.0	56.3	52.6	(29)
(30)			MCDWB	53.1	53.5	59.7	68.8	73.2	80.0	83.9	83.1	75.6	67.7	58.3	54.1	(30)
(31)		5%	WB	49.4	49.2	52.3	56.0	61.8	66.7	67.5	67.3	63.4	60.3	54.1	50.7	(31)
(32)			MCDWB	51.1	51.6	57.3	64.3	70.2	76.2	80.4	79.3	72.3	64.7	56.0	52.3	(32)
(33)		10%	WB	47.2	47.4	50.5	53.7	59.5	64.5	65.8	65.5	61.6	58.5	52.1	48.7	(33)
(34)			MCDWB	48.9	49.8	54.9	60.5	66.9	73.5	77.1	76.1	68.6	61.9	54.0	50.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	11.7	15.3	18.3	17.9	18.9	19.8	20.1	18.8	14.8	10.5	9.0	(35)
(36)			MCDBR	10.0	14.6	22.1	26.0	24.4	25.9	27.6	29.0	26.2	19.0	12.0	9.5	(36)
(37)			MCWBR	8.5	10.7	13.2	13.9	12.7	12.3	10.4	11.0	12.3	11.1	9.2	8.3	(37)
(38)		5% WB	MCDBR	9.7	13.1	17.7	23.3	21.5	22.9	23.6	23.8	21.3	16.5	10.8	9.3	(38)
(39)			MCWBR	8.7	10.4	11.5	13.1	12.2	12.0	9.8	10.4	10.8	10.3	8.9	8.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.322	0.342	0.373	0.397	0.403	0.407	0.404	0.394	0.368	0.359	0.333	0.317	(40)	
(41)		taud	2.425	2.379	2.287	2.223	2.246	2.272	2.275	2.323	2.380	2.409	2.432	2.397	(41)	
(42)		Ebn at Noon	235	256	265	270	272	272	270	268	264	246	228	219	(42)	
(43)		Edh at Noon	21	27	35	41	42	41	40	37	32	26	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	301	552	916	1387	1634	1828	1736	1514	1180	690	368	259	(44)	
(45)		RadStd	30	71	119	156	144	181	130	145	87	59	34	34	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page